

JEE MAIN REVISION MASTER: 3D GEOMETRY SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 385 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Planes & Line-Plane Intersections in 3D	209 PYQs	3D Planes
Very High	Direction Cosines & Equations of Lines in 3D	185 PYQs	3D Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Planes & Line-Plane Intersections in 3D	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, $d = \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$, $\sin\theta = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$
Direction Cosines & Equations of Lines in 3D	$l^2 + m^2 + n^2 = 1$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$, $d_{\text{skew}} = \frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Planes & Line-Plane Intersections in 3D (Very High Yield — 209 PYQs)

Core Principle: Normal vector to a plane, perpendicular distance from point to plane, and angle between a line and plane (uses $\sin\theta$).

■■ **Caution / JEE Pitfall:** Angle between a LINE (vector $\vec{b}$) and a PLANE (normal $\vec{n}$) is computed with SINE ($\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$), NOT cosine!

■ **Speed Trick:** A line is parallel to a plane if and only if it is perpendicular to the plane's normal: $\vec{b} \cdot \vec{n} = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $(x-4)/1 = (y-9)/2 = (z-5)/1$ be the point $(\lambda, \mu, 2)$." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let a triangle PQR be such that P and Q lie on the line $(x+3)/8 = (y-4)/2 = (z+1)/2$ and are at a distance of 6 units from R..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "The shortest distance between the lines $\vec{r} = ((1/3)\hat{i} + 2\hat{j} + 8\hat{k}) + \lambda(2\hat{i} - 5\hat{j} + 6\hat{k})$ and $\vec{r} = ((1/3)\hat{i} + 2\hat{j} + 8\hat{k}) + \mu(2\hat{i} - 5\hat{j} + 6\hat{k})$ is..." [Ans: B]

3.2 Direction Cosines & Equations of Lines in 3D (Very High Yield — 185 PYQs)

Core Principle: Direction cosines (l, m, n), line passing through two points, and shortest distance formula between skew lines.

■ **Caution / JEE Pitfall:** Direction ratios (a, b, c) can be any proportional numbers, but direction cosines (l, m, n) MUST satisfy $l^2 + m^2 + n^2 = 1$.

■ **Speed Trick:** Two lines in 3D intersect (are coplanar) if and only if the shortest distance between them is ZERO: $(\text{vec}(a)_2 - \text{vec}(a)_1) \cdot (\text{vec}(b)_1 \times \text{vec}(b)_2) = 0$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Let the point A be the foot of perpendicular drawn from the point P ($a, b, 0$) on the line $(x-1)/2 = (y-2)/1 = (z-\alpha)/3$" [Ans: N/A]
- **[2026 Evening Shift] Q#2 (MCQ):** "The shortest distance between the lines $(x-4)/1 = (y-3)/2 = (z-2)/-3$ and $(x+2)/2 = (y-6)/4 = (z-5)/-5$ is:..." [Ans: N/A]
- **[2026 Evening Shift] Q#2 (MCQ):** "If the distance of the point ($a, 2, 5$) from the image of the point ($1, 2, 7$) in the line $(x-1)/1 = (y-1)/1 = (z-2)/2$ is 4, then..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let a triangle PQR be such that P and Q lie on the line $(x+3)/8 = (y-4)/2 = (z+1)/2$ and are at a distance of 6 units from R(1, 2, 3). If (α, β, γ) is the centroid of ΔPQR , then $\alpha + \beta + \gamma$ is equal to :

- (A) 4
- (B) 5
- (C) 6 [Correct]
- (D) 8

Correct Answer: C

Solution Analysis: Write the given line in parametric form. $(x+3)/8 = (y-4)/2 = (z+1)/2 = t$ So any point on the line is $P(t) = (-3+8t, 4+2t, -1+2t)$ Since both P and Q lie on this line and are at a distance 6 from R(1, 2, 3), we find the points on the line whose distance from R is 6. Distance condition: $\sqrt{(x-1)^2 + (y-2)^2 + (z-3)^2} = 6$ Substitute $x = -3+8t, y = 4+2t, z = -1+2t$: $(-3+8t-1)^2 + (4+2t-2)^2 + (-1+2t-3)^2 = 36$ $(8t-4)^2 + (2t+2)^2 + (2t-4)^2 = 36$ Now expand: $64t^2 - 64t + 16 + 4t^2 + 8t + 4 + 4t^2 - 16t + 16 = 36$ $72t^2 - 72t + 36 = 36$ $72t^2 - 72t = 0$ $72t(t-1) = 0$ So, $t = 0$ or $t = 1$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#2)

If the distance of the point $(a, 2, 5)$ from the image of the point $(1, 2, 7)$ in the line $(x - 1) = (y - 1) = (z - 2) / 2$ is 4, then the sum of all possible values of a is equal to :

- (A) 11
- (B) 9
- (C) 6 **[Correct]**
- (D) 4

Correct Answer: C

Solution Analysis: Let the point $P(1, 2, 7)$. We need the image of P in the line $(x - 1) = (y - 1) = (z - 2) / 2$. This means the line passes through the point $A(0, 1, 2)$ and has direction ratios $\text{vec}(d) = (1, 1, 2)$. If P' is the image of P in this line, then the given line is the perpendicular bisector of the segment joining P and P' in the sense of reflection about a line in 3D. So first we find the foot of perpendicular from P to the line. Let the foot be M . A general point on the line is $M(t) = (t, 1+t, 2+2t)$. Now, $\text{vec}(PM) = (t-1, t-1, 2t-5)$. Since PM is perpendicular to the line, we